Artificial Neural Networks for Solving Ordinary and Partial Differential Equations

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Motivation

- Classical methods such as finite differences and finite elements are usually mesh-based.
- This paper proposes a different view: approximate the solution by a neural network.
- The differential equation is transformed into an optimization problem.

Differential equation + BC/IC

$\downarrow$

trial solution

$\downarrow$

residual minimization

$\downarrow$

approximate solution

Main idea

The neural network is used as a global function approximator for the unknown solution.

Main Methodology: Trial Solution

The approximate solution is not taken as the raw neural network output.

Instead, the paper constructs a **trial solution**:

$$\Psi_t(x) = A(x) + F\big(x, N(x, \mathbf{p})\big),$$

where

- $N(x, \mathbf{p})$ is the neural network,
- $A(x)$ is chosen to satisfy the boundary or initial conditions,
- F is designed so that the neural correction does not violate the constraints.

Key point

Boundary or initial conditions are satisfied **exactly by construction**.

Example: Trial Solutions for ODE

First-order ODE with initial condition

$$\frac{d\Psi(x)}{dx} = f(x, \Psi), \qquad \Psi(0) = A$$

A hard-constrained trial solution is chosen as

$$\Psi_t(x) = A + x\, N(x; \theta)$$

Since the correction term is multiplied by x, it vanishes at $x = 0$, so the initial condition is satisfied exactly.

Second-order ODE

$$\frac{d^2\Psi(x)}{dx^2} = f(x, \Psi, \Psi')$$

For an initial value problem:

$$\Psi(0) = A, \qquad \Psi'(0) = A'$$

the trial solution is

$$\Psi_t(x) = A + A'x + x^2 N(x; \theta)$$

For a two-point Dirichlet boundary condition:

$$\Psi(0) = A, \qquad \Psi(1) = B$$

the trial solution is

$$\Psi_t(x) = A(1 - x) + Bx + x(1 - x)N(x; \theta)$$

Key idea: the factors x, x^2, and $x(1 - x)$ are designed so that the neural correction vanishes at the constrained points, ensuring exact satisfaction of ICs/BCs by construction.

Example: Trial Solutions for PDE

Consider the Poisson equation on the unit square:

$$\frac{\partial^2 \Psi(x,y)}{\partial x^2} + \frac{\partial^2 \Psi(x,y)}{\partial y^2} = f(x,y), \qquad (x,y) \in [0,1]^2$$

with Dirichlet boundary conditions

$$\Psi(0,y) = f_0(y), \quad \Psi(1,y) = f_1(y), \quad \Psi(x,0) = g_0(x), \quad \Psi(x,1) = g_1(x).$$

The trial solution is chosen as

$$\Psi_t(x,y) = A(x,y) + x(1-x)y(1-y)\, N(x,y;\mathbf{p}).$$

$$x(1-x)y(1-y) = 0 \qquad \text{on the whole boundary.}$$

Interpretation

The neural network only learns an interior correction term, while $A(x,y)$ enforces the boundary conditions.

How is $A(x, y)$ constructed?

The function $A(x, y)$ is chosen so that the boundary conditions are satisfied exactly:

$$\Psi(0, y) = f_0(y), \quad \Psi(1, y) = f_1(y), \quad \Psi(x, 0) = g_0(x), \quad \Psi(x, 1) = g_1(x).$$

In the paper, it is defined as

$$\begin{aligned} A(x, y) = (1 - x) f_0(y) + x f_1(y) \\ + (1 - y)\Big[g_0(x) - \big((1 - x) g_0(0) + x g_0(1)\big)\Big] \\ + y\Big[g_1(x) - \big((1 - x) g_1(0) + x g_1(1)\big)\Big]. \end{aligned}$$

Interpretation

This construction combines the four boundary functions while correcting the corner contributions, so that the boundary values are matched consistently on all edges.

Training Principle and Our Reproduction

After constructing the trial solution, the residual is minimized on collocation points:

$$\mathcal{L}(\mathbf{p}) = \sum_i \left(\mathcal{D}[\Psi_t](x_i) - f(x_i)\right)^2 .$$

In our reproduction:

- derivatives are computed using **PyTorch automatic differentiation**,
- the training is done by residual minimization,
- several examples from the paper are reproduced.

Reproduced examples:

- first-order ODE,
- second-order boundary value problem,
- coupled ODE system,
- 2D PDE with comparison to FEM.

Example 1: First-Order ODE

We reproduce the paper's first ODE example:

$$\Psi'(x) + \left(x + \frac{1+3x^2}{1+x+x^3}\right)\Psi(x) = x^3 + 2x + x^2\frac{1+3x^2}{1+x+x^3}, \qquad \Psi(0) = 1, \quad x \in [0,1].$$

Its exact solution is

$$\Psi_a(x) = x^2 + \frac{e^{-x^2/2}}{1+x+x^3}.$$

Following the paper, we use the hard-constrained trial solution

$$\Psi_t(x) = 1 + x\,N(x;\theta).$$

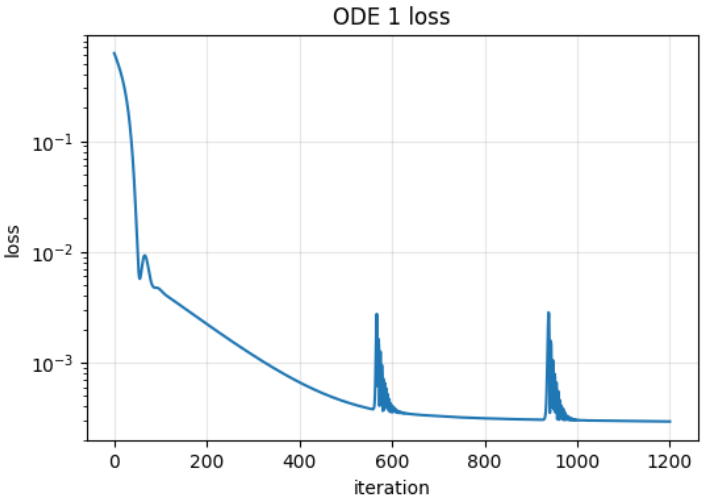
Key point

Since the correction term is multiplied by x, it vanishes at $x = 0$. Therefore, the initial condition is satisfied exactly:

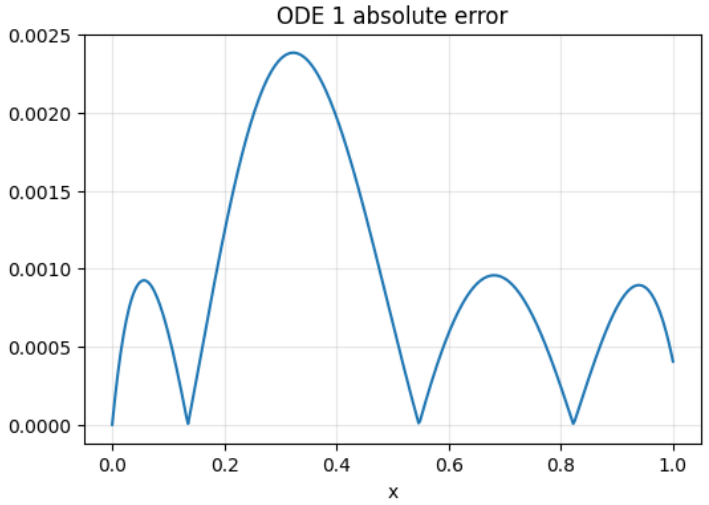
$$\Psi_t(0) = 1.$$

Example 1: Results

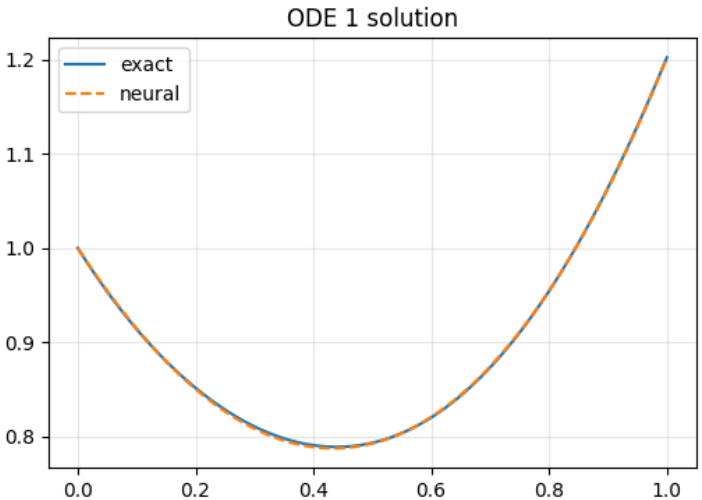
Training loss



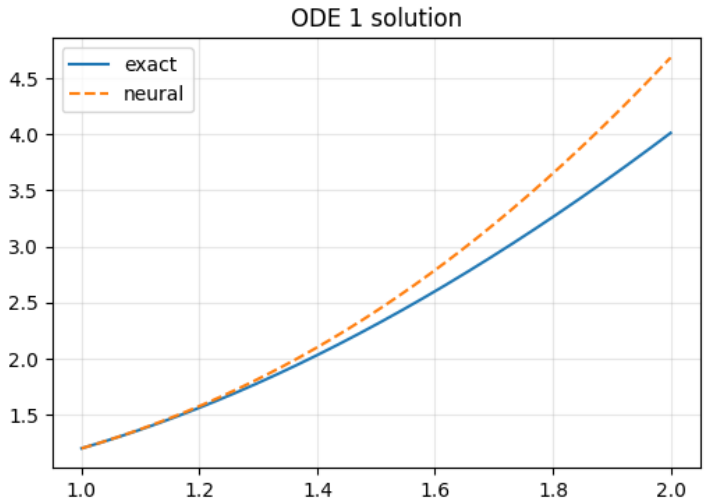
Pointwise error



In-domain fitting



Extrapolation



Example 2: Second-Order ODE with Boundary Conditions

We reproduce the second ODE example:

$$\Psi''(x) + \frac{1}{5}\Psi'(x) + \Psi(x) = -\frac{1}{5}e^{-x/5}\cos x, \quad x \in [0, 1].$$

The boundary conditions are

$$\Psi(0) = 0, \qquad \Psi(1) = \sin(1)e^{-1/5}.$$

The exact solution is

$$\Psi_a(x) = e^{-x/5}\sin x.$$

Following the paper, we use the hard-constrained trial solution

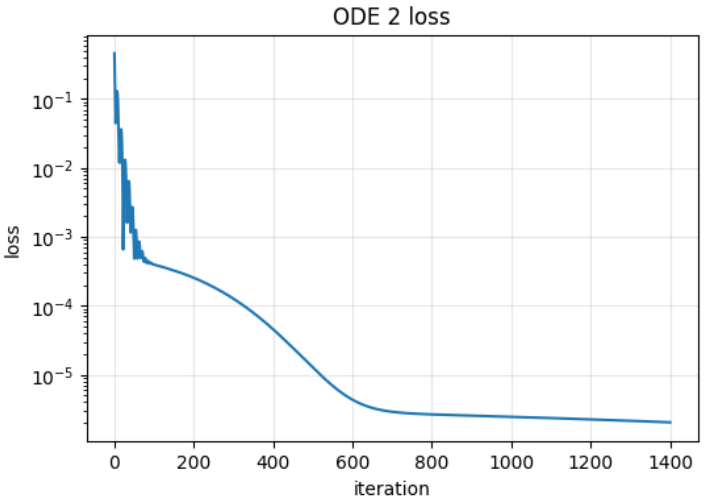
$$\Psi_t(x) = x\,\Psi(1) + x(1-x)N(x;\theta).$$

Key point

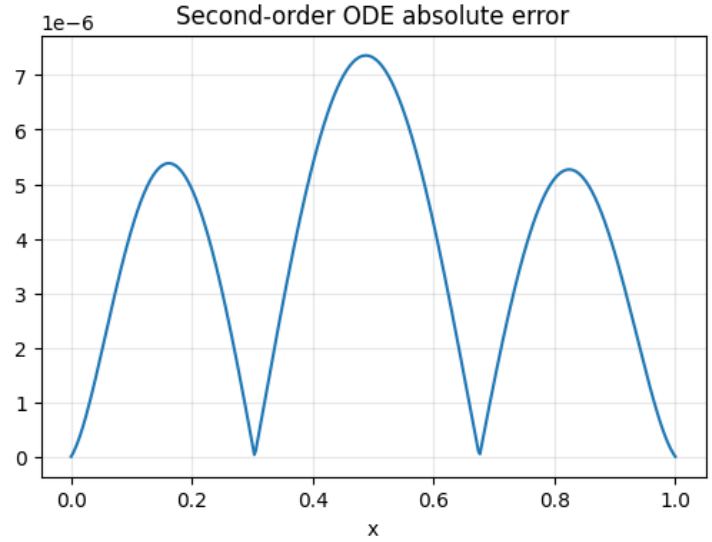
Since $x(1-x) = 0$ at both endpoints, the neural correction vanishes at $x = 0$ and $x = 1$. Therefore, the boundary conditions are satisfied exactly.

Example 2: Results

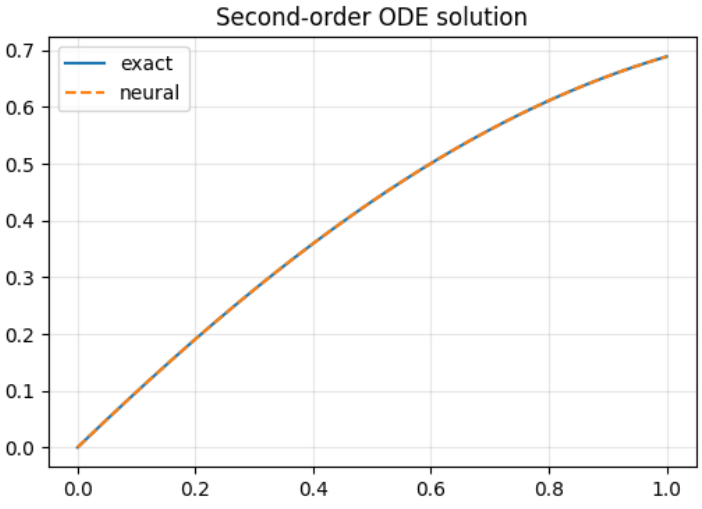
Training loss



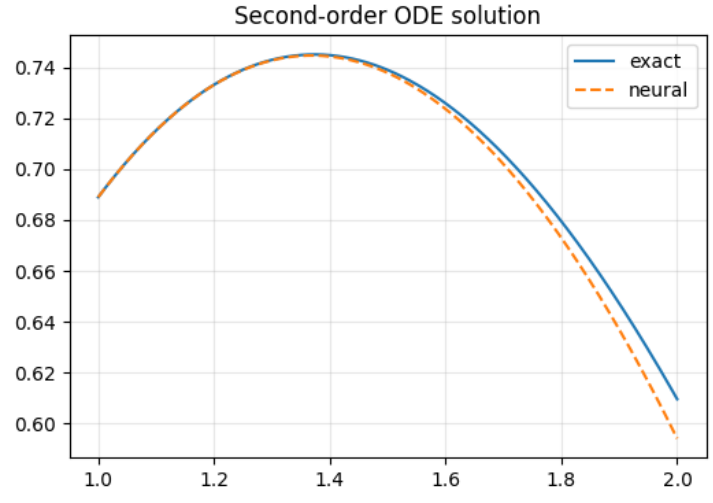
Pointwise error



In-domain fitting



Extrapolation



Example 3: Coupled First-Order ODE System

Differential system

$$\Psi_1'(x) = \cos x + \Psi_1(x)^2 + \Psi_2(x) - \left(1 + x^2 + \sin^2 x\right),$$

$$\Psi_2'(x) = 2x - (1 + x^2)\sin x + \Psi_1(x)\Psi_2(x), \quad x \in [0, 3]$$

Initial conditions

$$\Psi_1(0) = 0, \qquad \Psi_2(0) = 1$$

Exact solutions

$$\Psi_1(x) = \sin x, \qquad \Psi_2(x) = 1 + x^2$$

Trial solutions

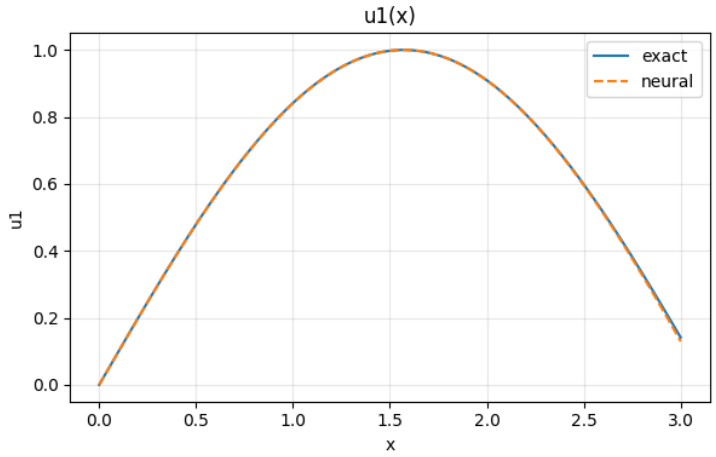
$$\Psi_{1,t}(x) = x\, N_1(x; \theta_1), \qquad \Psi_{2,t}(x) = 1 + x\, N_2(x; \theta_2)$$

Why this form?

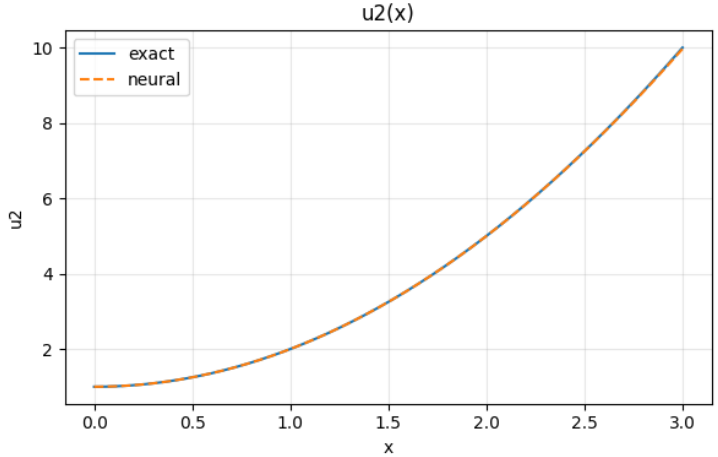
The factor x makes the neural correction vanish at $x = 0$, so the initial conditions are enforced exactly for both unknown functions.

Example 3: Results

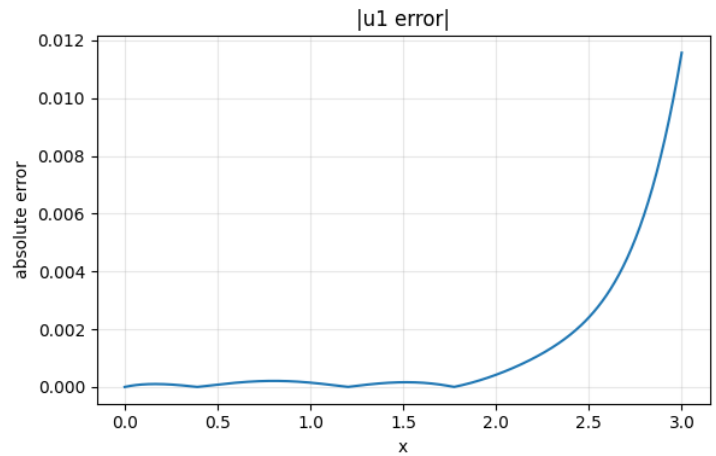
u1(x)



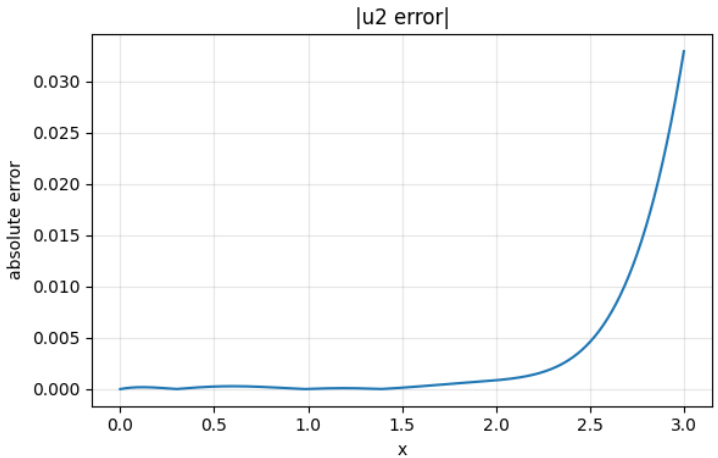
u2(x)



Absolute error u1



Absolute error u2



Example 4: 2D Poisson PDE with Dirichlet Conditions

Partial differential equation

$$\nabla^2 \Psi(x, y) = e^{-x}(x - 2 + y^3 + 6y), \qquad (x, y) \in [0, 1]^2$$

Dirichlet boundary conditions

$$\Psi(0, y) = y^3, \qquad \Psi(1, y) = (1 + y^3)e^{-1},$$
$$\Psi(x, 0) = xe^{-x}, \qquad \Psi(x, 1) = e^{-x}(x + 1)$$

Exact solution

$$\Psi_a(x, y) = e^{-x}(x + y^3)$$

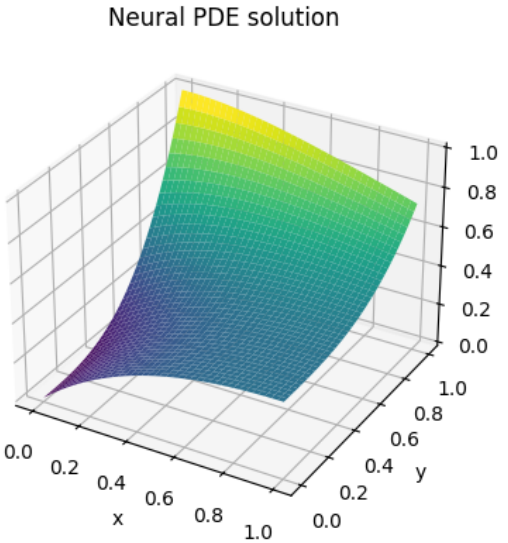
Trial solution

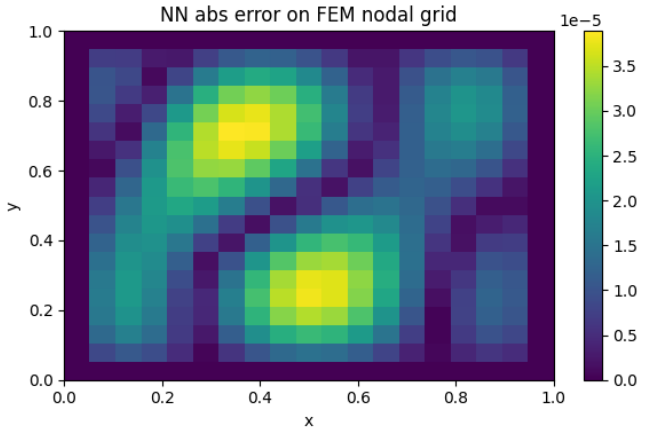
$$\Psi_t(x, y) = A(x, y) + x(1 - x)y(1 - y)N(x, y; \theta)$$

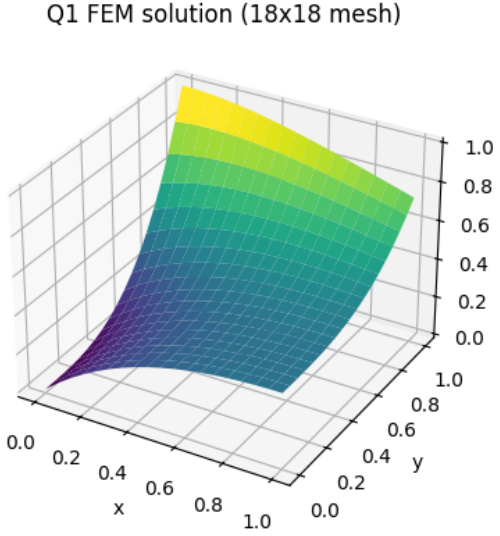
Why this form?

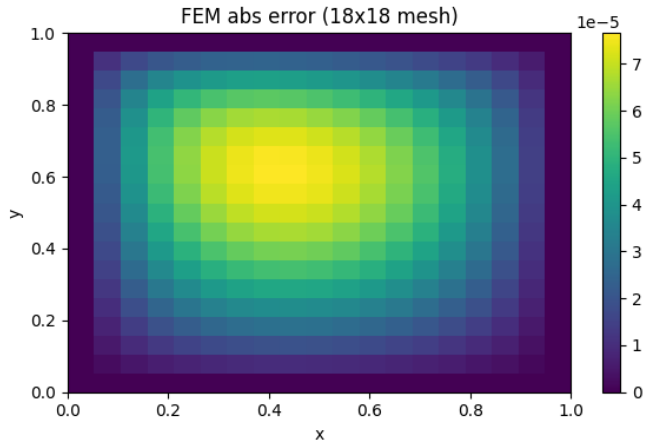
Since $x(1 - x)y(1 - y) = 0$ on the whole boundary of the square, the neural correction vanishes there. Thus, the boundary values are entirely determined by $A(x, y)$.

Example 4: ANN and FEM Comparison

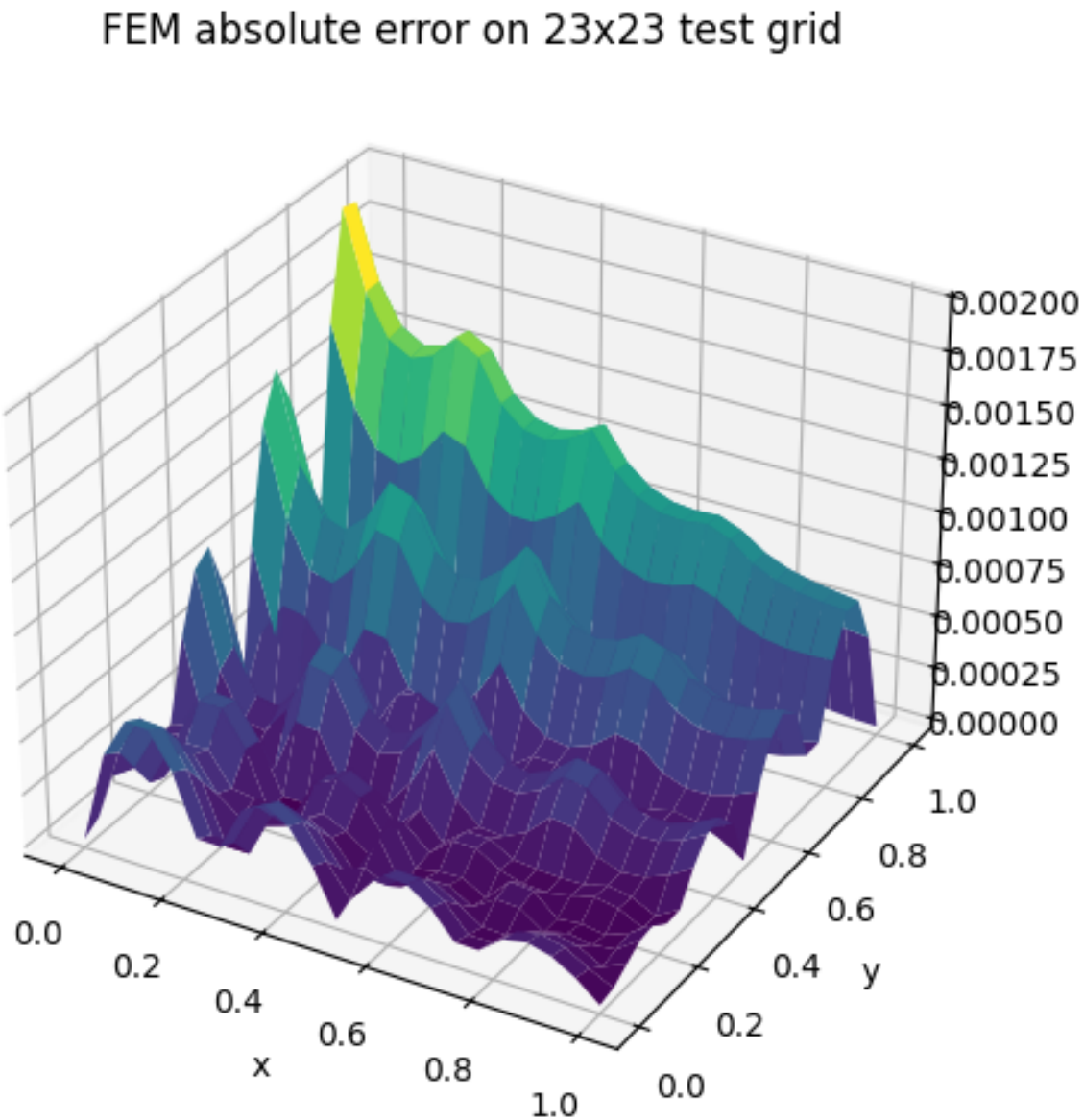








Example 4: ANN and FEM Comparison



Conclusion

- The trial-solution construction is the core idea of the paper.
- It enforces initial or boundary conditions exactly.
- Our reproduction confirms strong in-domain approximation for ODEs and PDEs.
- Extrapolation is weaker than interpolation.
- For the PDE example, FEM remains a strong classical baseline.

Take-home message

This paper provides an elegant neural formulation for solving differential equations and can be viewed as an early precursor of modern PINN-style methods.

Thank you for your attention.